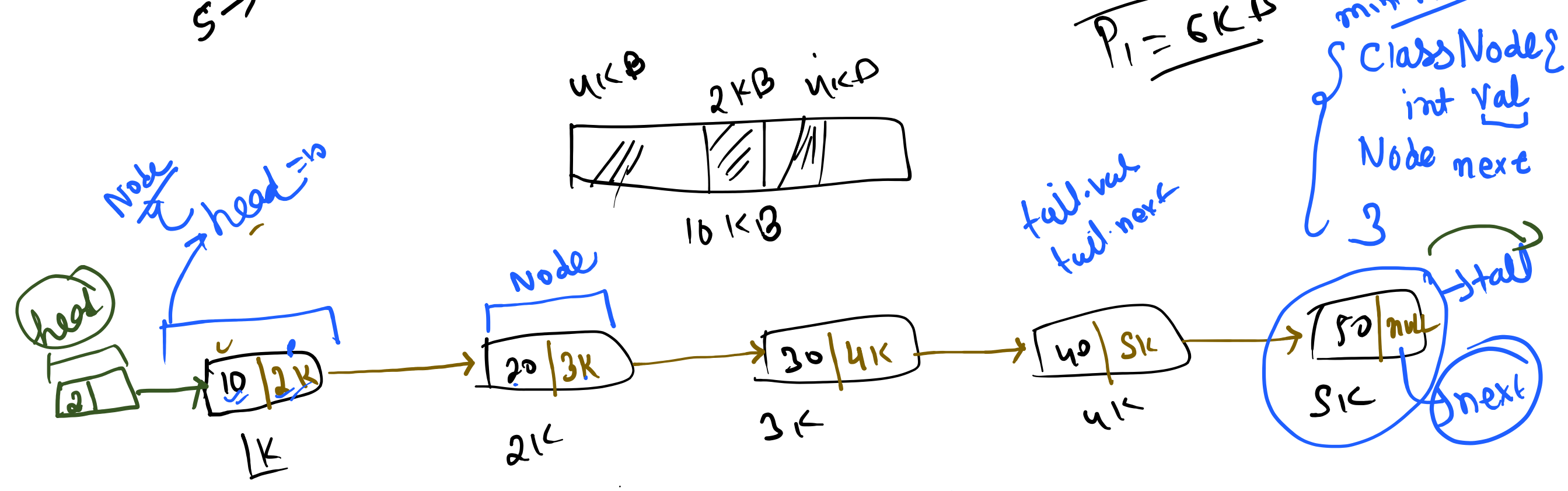
Lec-27
15/02/26 12:51 PM

LinkedList → non- cont

S → 20B

P1 = 6KB

mini mn
class Node{
int val
Node next
}


4KB
2KB 4KB
10KB
Node
head=10
head
Node
10 | 2K
1K
20 | 3K
2K
30 | 4K
3K
40 | 5K
4K
50 | null
5K
tail.val
tail.next
tail
next

Student s = new Student()
S.name = Raj.

name = null
age = 0

class Student{
String name
int age
}

(i) AddFirst
AddLast
AddatIndex

(ii) remove first
remove Last
rem at index

(iii) getFirst
get last
get at index

(iv) display

(v) size

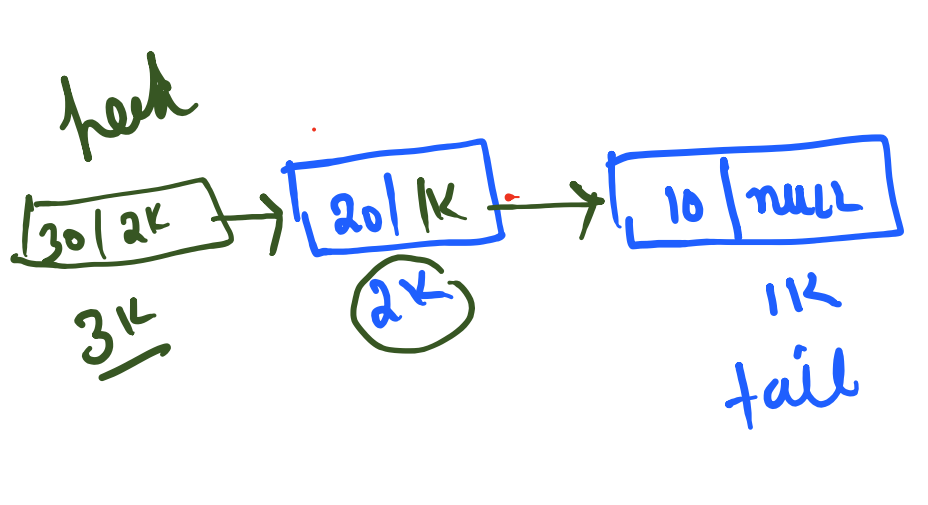
nn.val = item

size = 0

ll.addF(10) ←
ll.addF(20)
ll.addF(30)

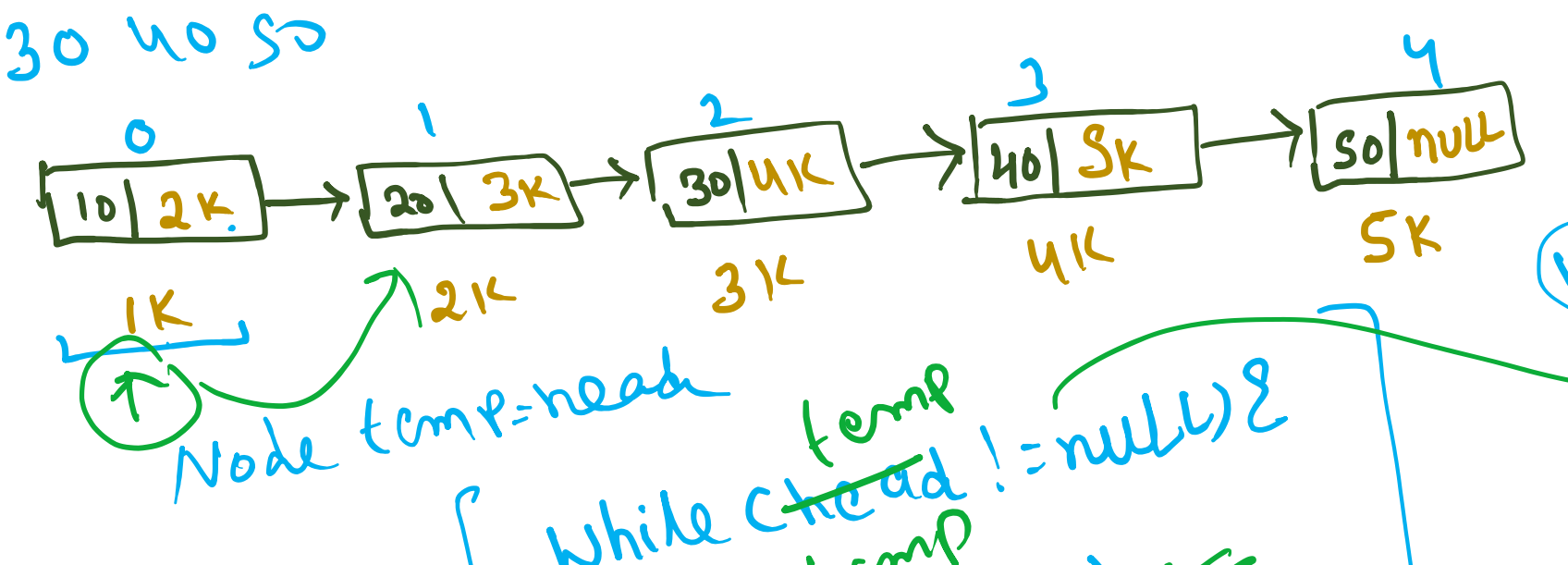
```
public void AddFirst(int item) {

}
```


head
30 | 2K
3K
20 | 1K
2K
10 | null
1K
tail

1K
```
Node nn = new Node(item)
if( size==0){
    head=nn
    tail=nn
    size++
}
else{
    nn.next = head
    head = nn
    size++
}
```

10 20 30 40 50


0
10 | 2K
1K
1
20 | 3K
2K
2
30 | 4K
3K
3
40 | 5K
4K
4
50 | null
5K

head=1K

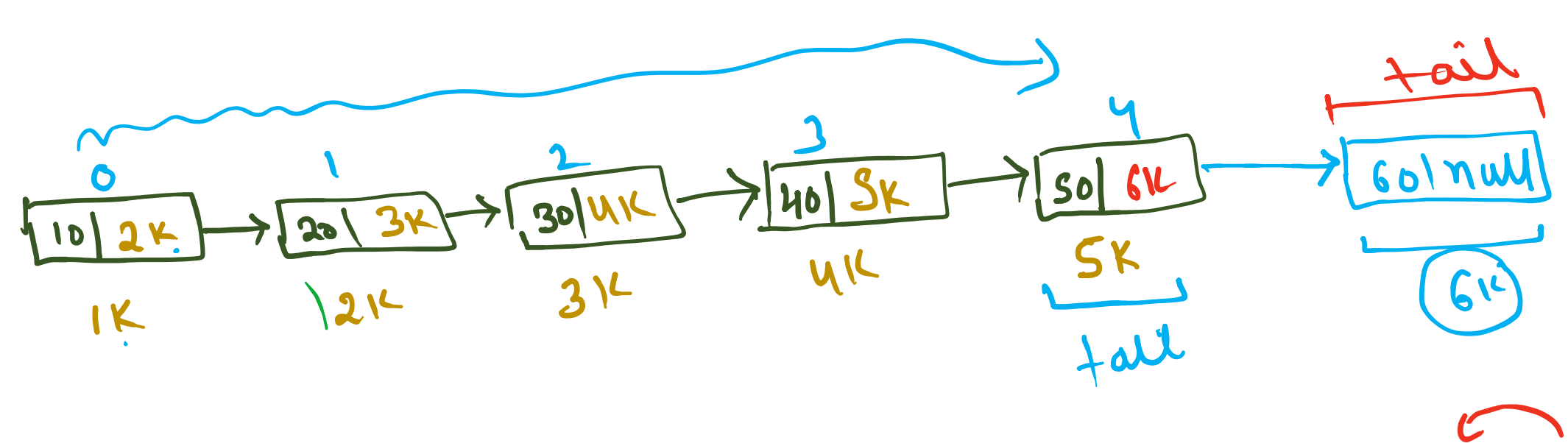
head = null

Node temp=head

```
While( ~~head~~ temp !=null){
    SySo( ~~head~~ temp.val)
    ~~head~~ temp = ~~head~~ temp.next
}
```

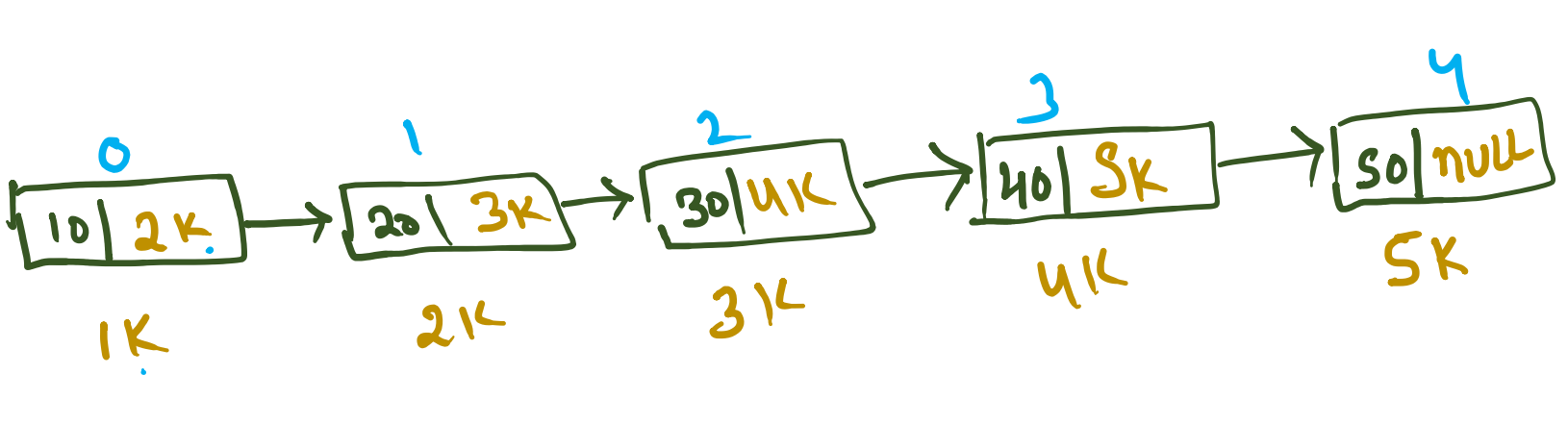
```
While(i<n){
    S.o.p(arr(i))
    i++
}
```

SIso(arr(i))
i++


0
10 | 2K
1K
1
20 | 3K
2K
2
30 | 4K
3K
3
40 | 5K
4K
4
50 | 6K
5K
tail
tail
60 | null
6K

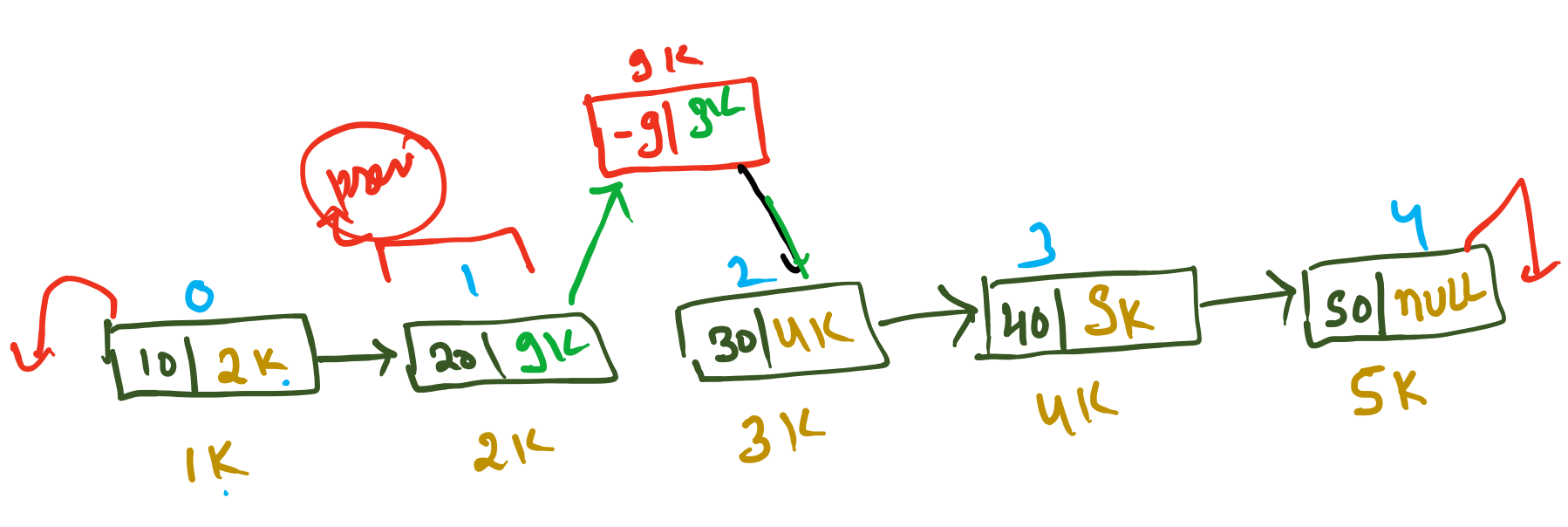
Node nn = new Node(item)
6K

```
{
    tail.next=nn
    tail=nn
    size++
}
```

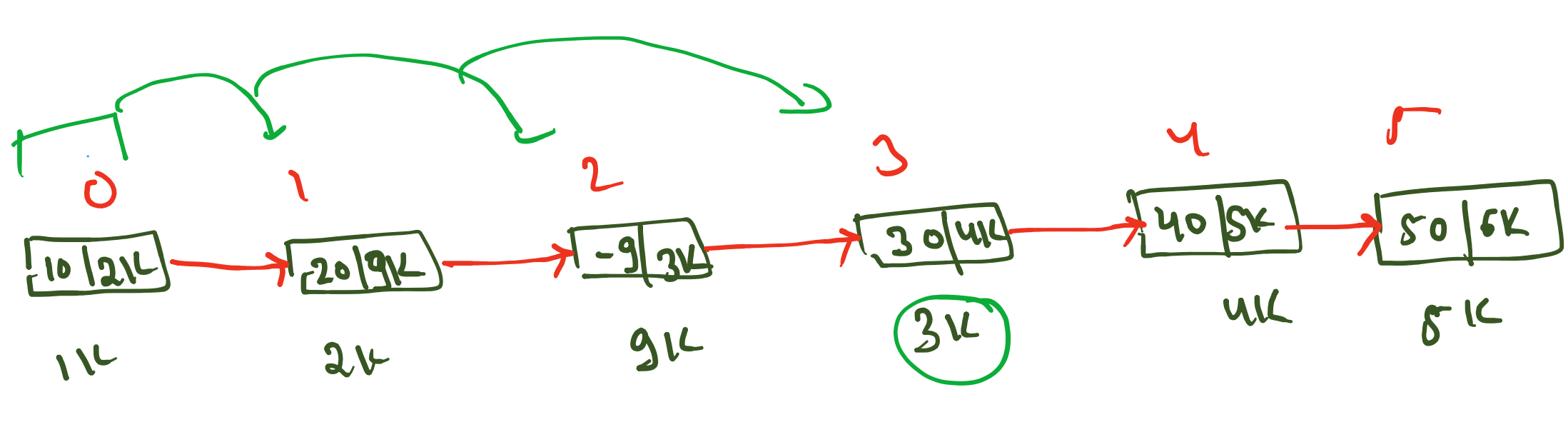

0
10 | 2K
1K
1
20 | 3K
2K
2
30 | 4K
3K
3
40 | 5K
4K
4
50 | null
5K

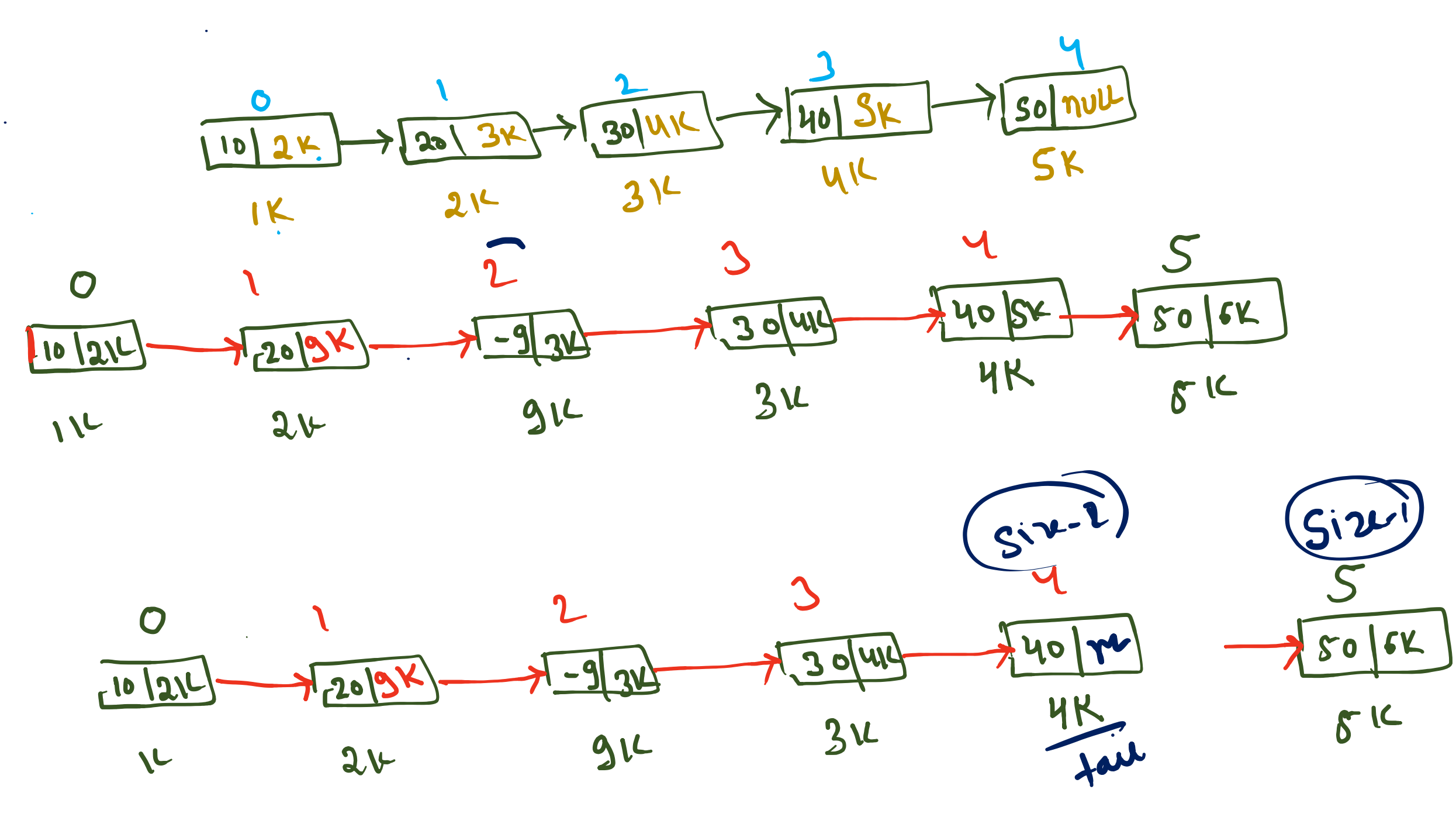
K=2

```
for(i=0 i<K i++){
    temp= temp.next
}
```

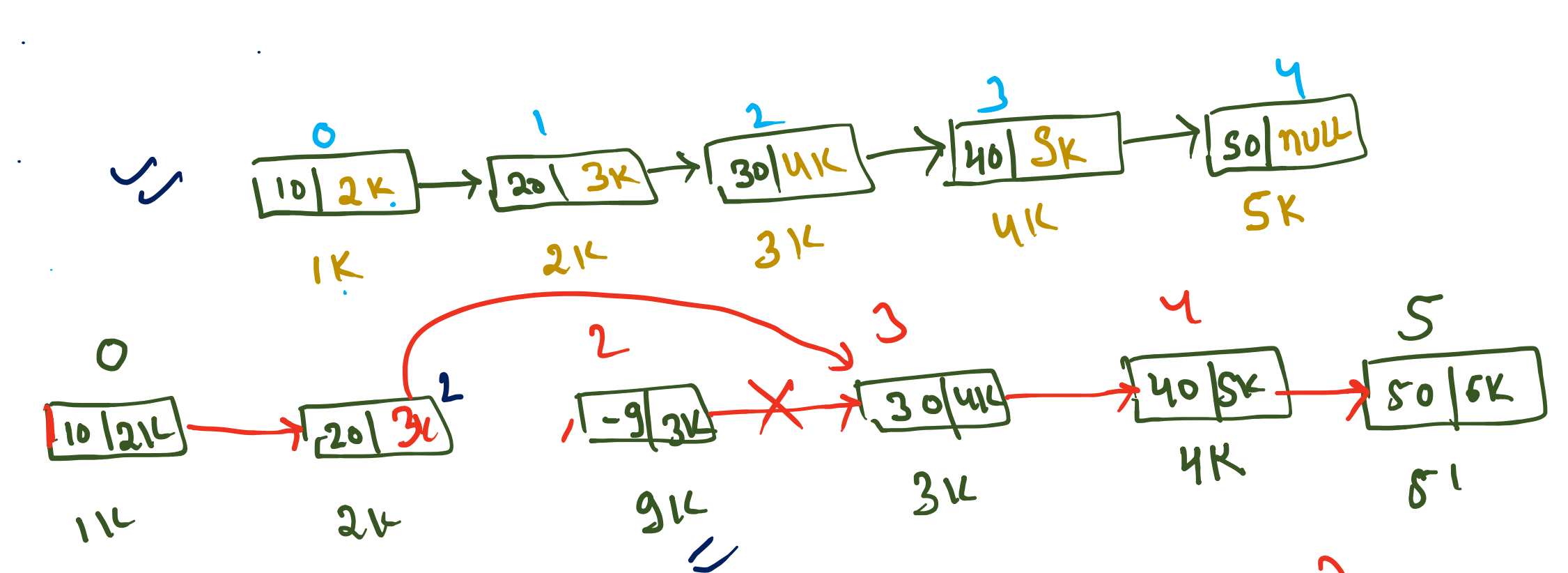

prev
9K
-9 | 3K
0
10 | 2K
1K
1
20 | 9K
2K
2
30 | 4K
3K
3
40 | 5K
4K
4
50 | null
5K

-9
3K
nn.next = prev.next
→ prev.next = nn

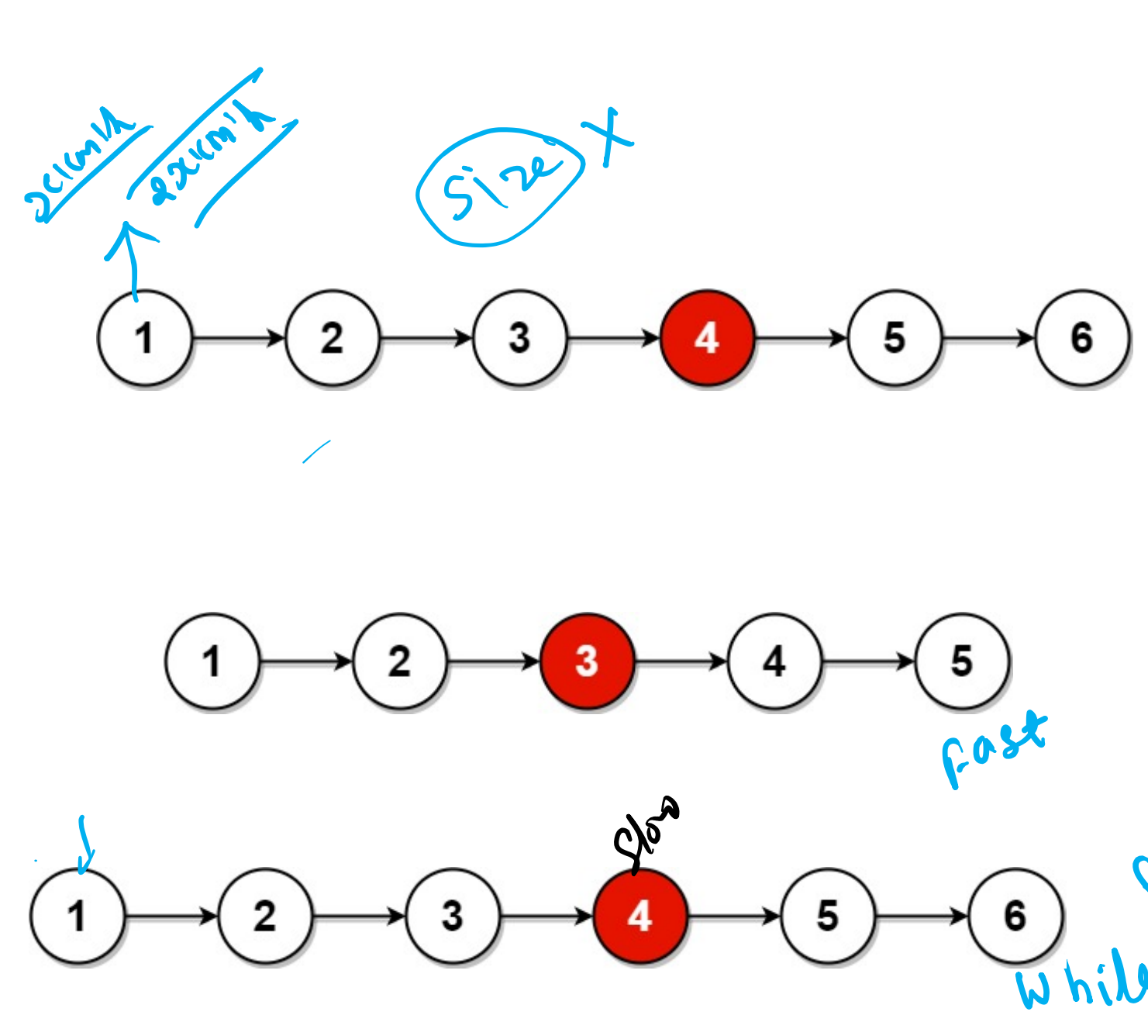

0
10 | 2K
1K
1
20 | 9K
2K
2
-9 | 3K
9K
3
30 | 4K
3K
4
40 | 5K
4K
5
50 | 6K
5K


0
10 | 2K
1K
1
20 | 3K
2K
2
30 | 4K
3K
3
40 | 5K
4K
4
50 | null
5K
0
10 | 2K
1K
1
20 | 9K
2K
2
-9 | 3K
9K
3
30 | 4K
3K
4
40 | 5K
4K
5
50 | 6K
5K
size-1
size-1
4K
tail

Node temp = head
head = head.next
temp.next = null


0
10 | 2K
1K
1
20 | 3K
2K
2
30 | 4K
3K
3
40 | 5K
4K
4
50 | null
5K
0
10 | 2K
1K
1
20 | 3K
2K
2
-9 | 3K
9K
3
30 | 4K
3K
4
40 | 5K
4K
5
50 | 6K
5K

```
Node prev = getNK(K-1)
Node curr = prev.next
prev.next = curr.next
curr.next = null
```

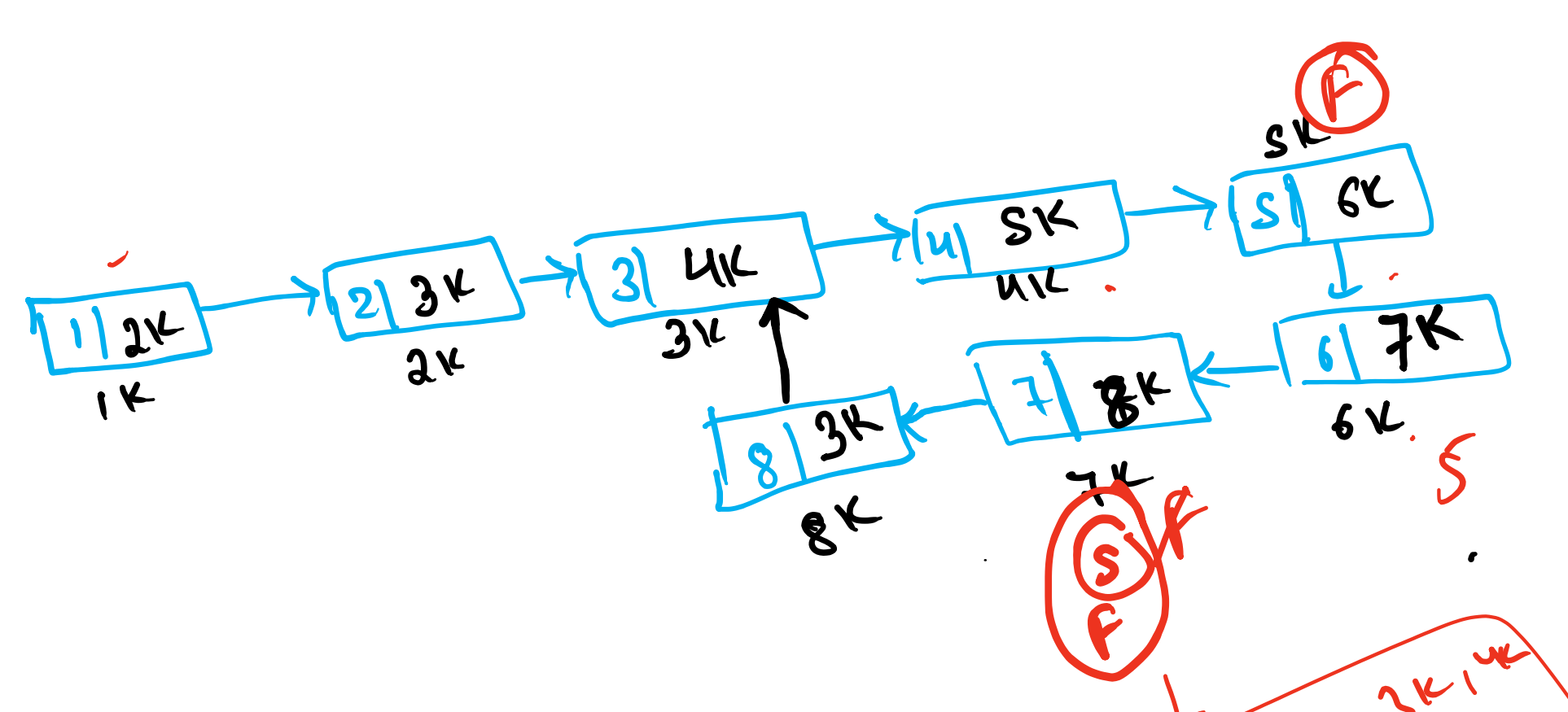

1
2
3
4
5
6
size
1
2
3
4
5
fast
slow
1
2
3
4
5
6

```
size=1   temp=head
While(temp!=null){
    size++
    temp=temp.next
}
```

5/2=2   6/2=3

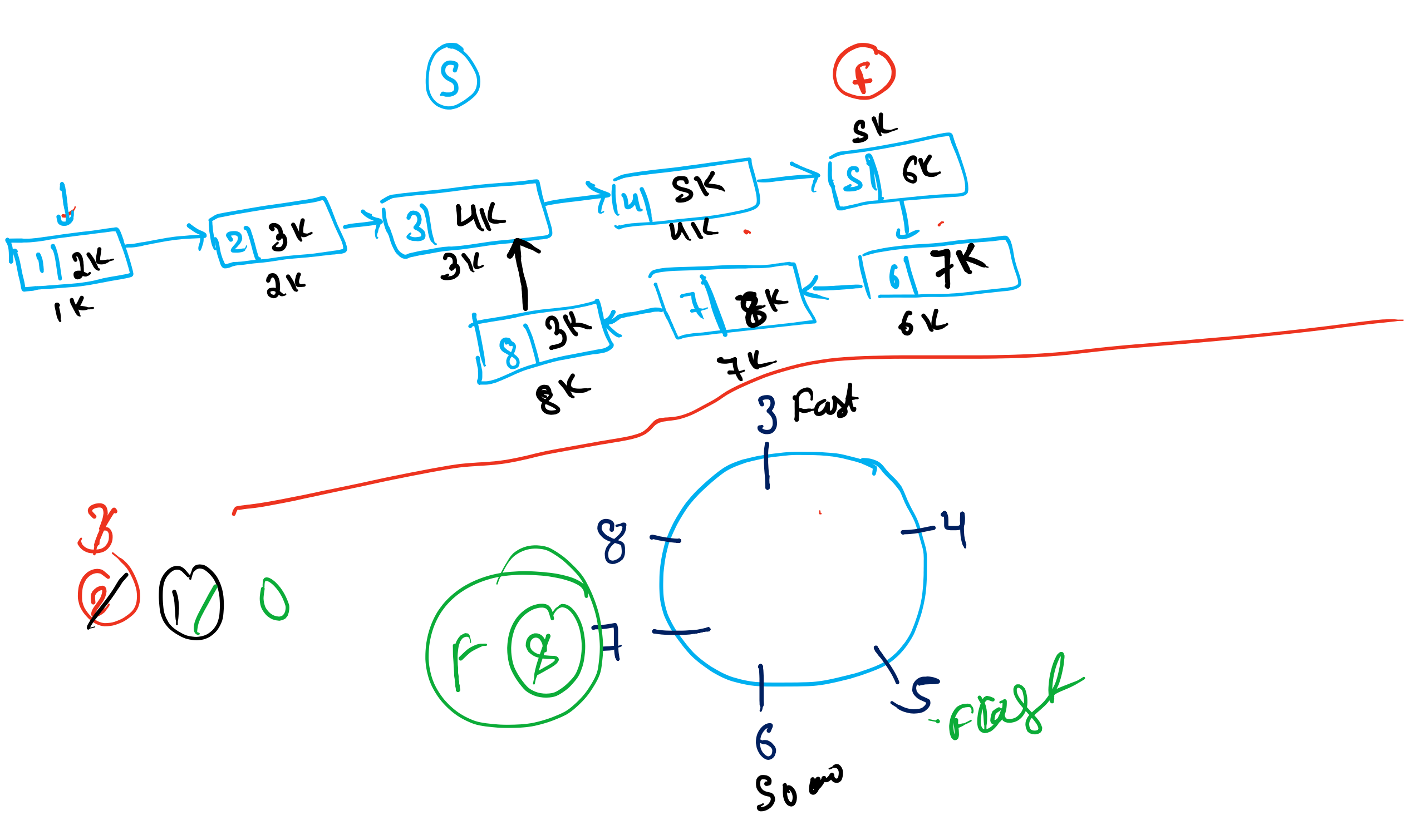
fast!=null

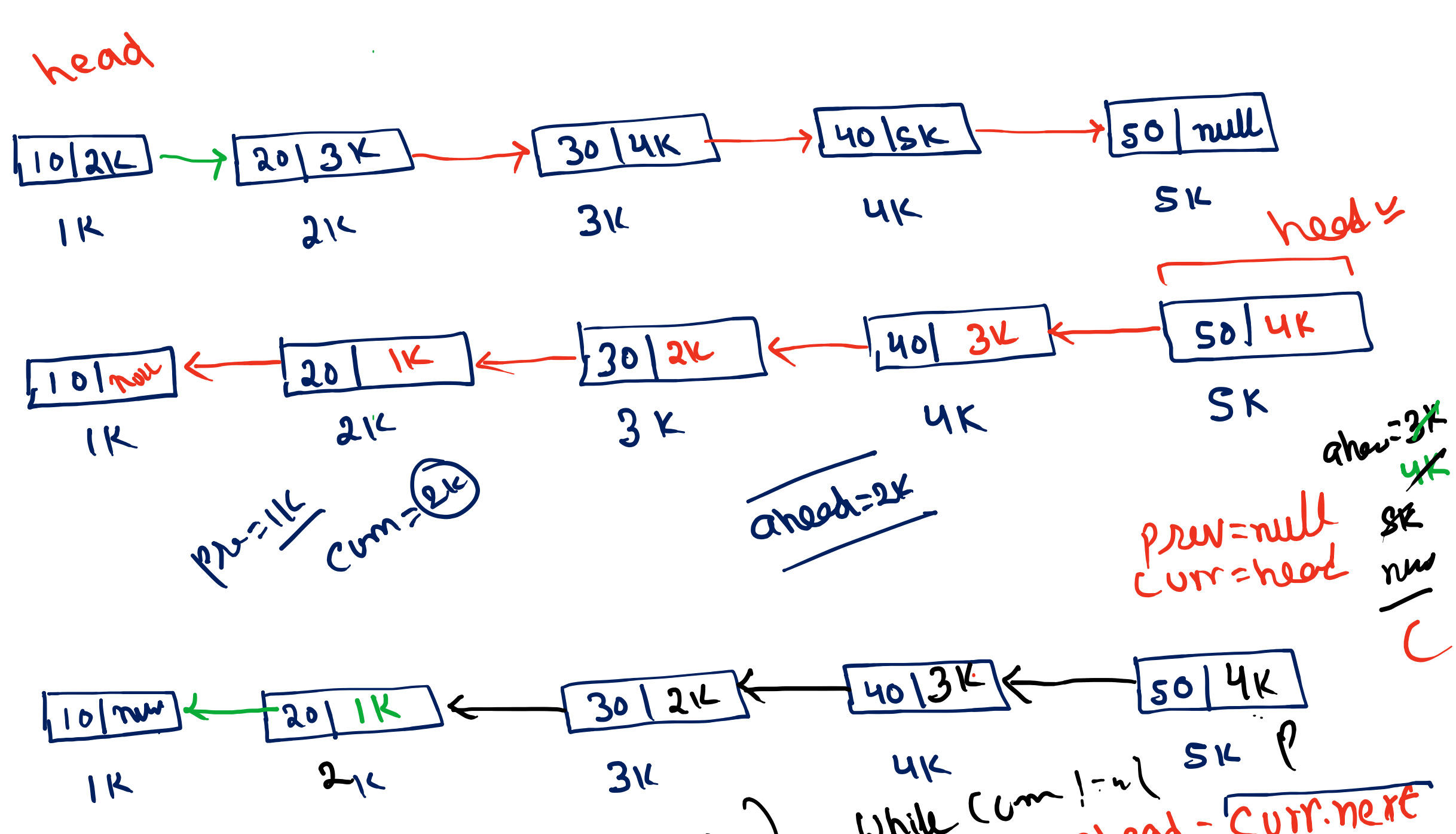
```
While( fast!=null && fast.next!=null ){
    slow=slow.next
    fast=fast.next.next
}
```


F
5K
1 | 2K
1K
2 | 3K
2K
3 | 4K
3K
4 | 5K
4K
5 | 6K
6 | 7K
6K
7 | 8K
7K
8 | 3K
8K
S
F
5

n²

1K, 2K, 3K, 4K, 5K, 6K, 7K, 8K


S
F
5K
1 | 2K
1K
2 | 3K
2K
3 | 4K
3K
4 | 5K
4K
5 | 6K
6 | 7K
6K
7 | 8K
7K
8 | 3K
8K
3 Fast
4
5 Fast
6 Slow
7
8
F
8
1
0


head
10 | 2K
1K
20 | 3K
2K
30 | 4K
3K
40 | 5K
4K
50 | null
5K
head
10 | null
1K
20 | 1K
2K
30 | 2K
3K
40 | 3K
4K
50 | 4K
5K
prev=1K
curr=2K
ahead=2K
ahead=~~3K~~ ~~4K~~ 5K null
10 | null
1K
20 | 1K
2K
30 | 2K
3K
40 | 3K
4K
50 | 4K
5K
p

prev=null
curr=head

O(N)

```
While(curr!=null){
    Node ahead = curr.next
    curr.next = prev
    prev = curr
    curr = ahead
}
```

prev